Connection and driver troubleshooting

Developer guide V0.1 | Source software v1.1.0 | 2026-10-11

Software and communication checks. Hardware repair, brake/steering calibration and missing fault-bit tables are outside this guide.

Status: Software reference

Separate the layers before changing settings

1. Confirm whether the session uses a simulator or physical CAN.
2. Check Linux adapter detection and interface state.
3. Confirm incoming feedback with a passive CAN capture.
4. Inspect the ROS node, topics and diagnostics.
5. Check the selected build/interface and only then review unit-specific parameters.

```
ip -br link
ip -details -statistics link show can0
candump can0
ros2 node list
ros2 topic echo --once /diagnostics
```

Common symptoms

Symptom	Check next
cannot open CAN interface	Adapter present? Correct can0/can1 name? Interface UP?
No feedback	Power, wiring, termination, confirmed bitrate and selected protocol
Feedback present, no motion	Approved operating mode, emergency stop, software state and command topic
Stops after about 0.5 s	Command publisher continuing? Source watchdog timeout? Do not mask the symptom by extending it first
command_limited	Requested motion exceeds software limits or confirmed steering range
ERROR-PASSIVE / BUS-OFF	Resolve wiring, termination or bitrate before sending commands
Unexpected wheel/lamp response on A	Firmware-dependent ID and wheel-order mapping; stop and confirm the mapping
Implausible battery readings	Raw value versus scaling and approved unit readings
Yaw drift	Wheel-odometry limitation; inspect sensor fusion and measured transforms

Keep B interfaces separate

Interface	Interpretation
legacy_b / VCU V2.6	Confirm frame ID radix, bitrate and steering sign; odometry feedback is coarse
Draft LCI	Heartbeat, checksum, live-limit and IMU diagnostics describe the design/simulator until firmware is confirmed

Scope note. Do not apply a draft-LCI maintenance reset to a VCU unit or assume a software estop service clears a physical fault. Use the approved operating procedure or contact support for a matching recovery procedure.

A VCU fault category is not the LCI severity level. This guide supplies no missing motor, EPS, EPB or BMS fault-bit definitions.

Collect a useful support record

Record the chassis build, interface, firmware identification if available, adapter/bitrate, software v1.1.0, exact symptom and timestamp. Save the driver log and a short passive capture.

```
ros2 topic echo --once /diagnostics
ros2 param dump /luchs_driver > luchs_driver_params.yaml
candump -l can0
# Stop the capture with Ctrl-C after about 10 seconds.
```

Review logs for your own project information before sharing them with support. Include raw and scaled values when reporting a decoding problem. This website update did not perform a hardware repair or a real-vehicle validation.

Source and edition

Prepared and reorganised by RJ Tech from the already-public Luchs ROS 2 v1.1.0 package by TerraAgent Robotics. Source licence: Apache-2.0. Source text and examples were shortened, explained and scoped for this topic. Original archives and protocol files remain unchanged.

Source file: docs/troubleshooting.md

Source file: docs/can_setup.md

Source file: luchs_driver/README.md

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